

Chapter 16

Support and Movement

Class XII

INTER KARACHI

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CHAPTER # 02: SUPPORT & MOVEMENT

SUPPORT AND MOVEMENT IN ANIMALS

In higher animals skeleton and muscles work together and help in locomotion. The muscles are attached to the skeleton. They both form a system, called **skeletal muscles system** or **musculo-skeletal system**. The nervous system co-ordinates these two system together.

HUMAN SKELETAL SYSTEM

In the body of animals there is a frame work called skeleton. A skeletal system usually consists of a single or a set of hard structure which provide the body with shape, support and protection and the power of movement.

ENDOSKELETON:

Occurrence:

- This type of skeleton lies inside the body, **Although** found in few invertebrates like spicules of sponges and **calcareous** plates of star fish.
- It is the only complex **skeleton** of vertebrates, which is a true **endoskeleton**.

Composition:

- The endoskeleton in the **vertebrates** is an internal frame **work** of the body being composed of two types of living (**rigid connective tissue**) materials, **the** bones and cartilages.

Significance:

- Cartilages and bones articulate with each other at a **variety** of joints that allows a wide range of movements brought about the **contraction** of **muscles** attached to them.
- As the endoskeleton of vertebrates is, **made up of living material** it grows in correlation with the growth of animal.

HUMAN SKELETON

Introduction:

- In human endoskeleton is **present** which consists of 206 bones are connected with each other by joints.

COMPOSITION OF HUMAN SKELETON

1. BONES:

Preface:

- Bone is the most rigid form of **connective tissues**.
- Bones form strong part of skeleton, these are present in limbs etc.
- These cells of bones are called as **osteocytes**.
- They also secrete jell like matrix around them which contain a network of collagen fiber.

Ossification:

- It is not soft like cartilage but it is **hardened** due to deposition of osteoblasts and crystals of calcium phosphate, this process is called as ossification or calcification.

Requirements:

- The process of ossification takes place in the presence of vitamin D which is necessary for Calcium absorption from the blood and milk.
- Milk, a major part of child's diet is a good source of Calcium. Thus proteins and mineral salts together make the bones hard and elastic.

Gross Anatomy of Bones

- A long bone has two main regions: the diaphysis and the epiphysis

- The diaphysis is the hollow, tubular shaft that runs between the proximal and distal ends of the bone. Inside the diaphysis is the medullary cavity, which is filled with yellow bone marrow in an adult.
- The outer walls of the diaphysis (cortex, cortical bone) are composed of dense and hard compact bone, a form of osseous tissue.
- The wider section at each end of the bone is called the **epiphysis** (plural = **epiphyses**), which is filled internally with spongy bone, another type of osseous tissue.
- Red bone marrow fills the spaces between the spongy bone in some long bones. Each epiphysis meets the diaphysis at the **metaphysis**.
- During growth, the metaphysis contains the **epiphyseal plate**, the site of long bone elongation. When the bone stops growing in **early adulthood** (approximately 18–21 years), the epiphyseal plate becomes an **epiphyseal line** seen in the figure.

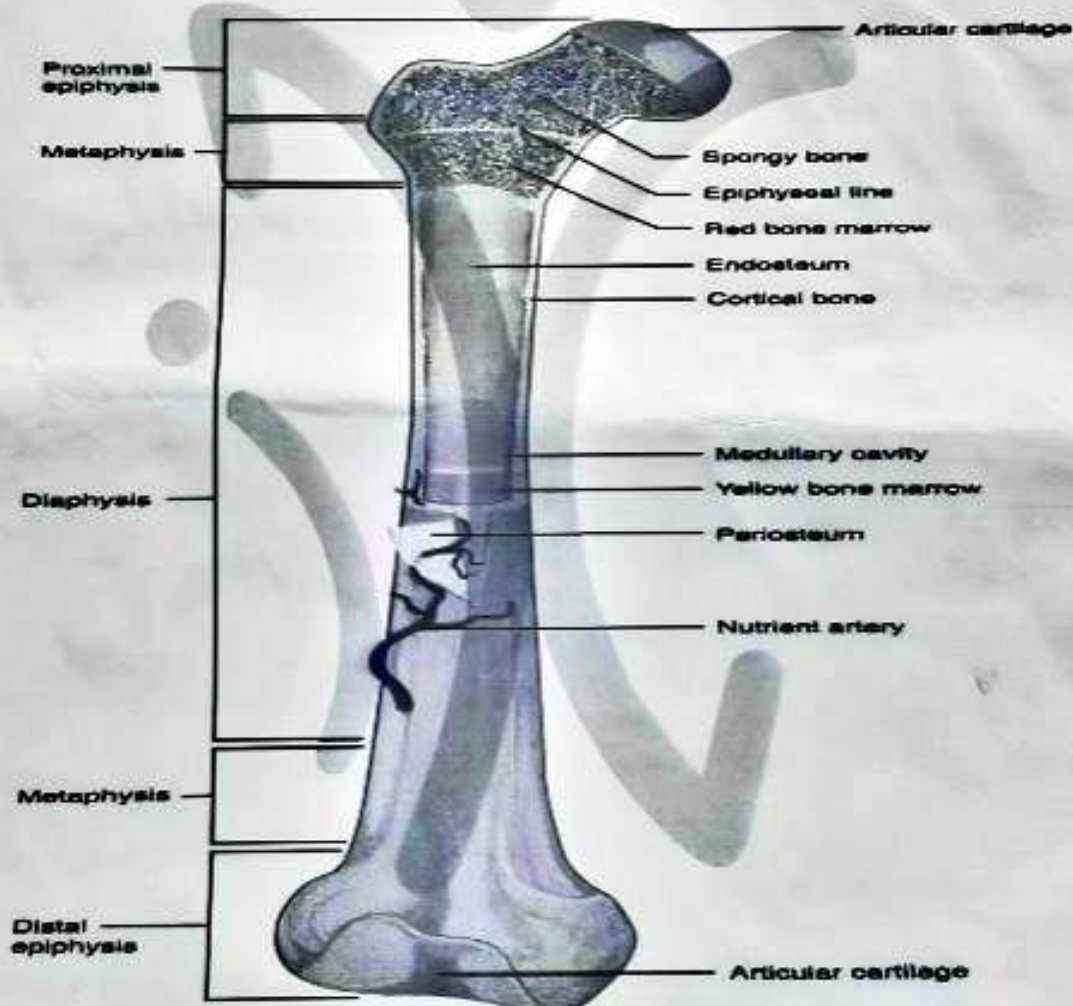


Figure 1 – Anatomy of a Long Bone: A typical long bone showing gross anatomical features.

Structure of Bone Tissue:

There are two types of bone tissue: compact and spongy. The names imply that the two types differ in density, or how tightly the tissue is packed together. There are three types of cells that contribute to bone homeostasis. Osteoblasts are bone-forming cells, osteoclasts resorb or break down bone, and osteocytes are mature bone cells. An equilibrium between osteoblasts and osteoclasts maintains bone tissue.

Compact Bone:

Compact bone consists of closely packed osteons or haversian systems. The osteon consists of a central canal called the osteonic (haversian) canal, which is surrounded by concentric rings (lamellae) of matrix. Between the rings of matrix, the bone cells (osteocytes) are located in spaces called lacunae. Small channels (canaliculi) radiate from the lacunae to the osteonic (haversian) canal to provide passageways through the hard matrix. In compact bone, the haversian systems are packed tightly together to form what appears to be a solid mass. The osteonic canals contain blood vessels that are parallel to the long axis of the bone. These blood vessels interconnect, by way of perforating canals, with vessels on the surface of the bone.

Spongy (Cancellous) Bone:

Spongy (cancellous) bone is lighter and less dense than compact bone. Spongy bone consists of plates (trabeculae) and bars of bone adjacent to small, irregular cavities that contain red bone marrow. The canaliculi connect to the adjacent cavities, instead of a central haversian canal, to receive their blood supply. It may appear that the trabeculae are arranged in a haphazard manner, but they are organized to provide maximum strength similar to braces that are used to support a building. The trabeculae of spongy bone follow the lines of stress and can realign if the direction of stress changes.

Bone Cells

Although bone cells compose less than 2% of the bone mass, they are crucial to the function of bones. Four types of cells are found within bone tissue: osteoblasts, osteocytes, osteogenic cells, and osteoclasts

Bone Cells		
Cell Type	Function	Location
Osteogenic cells	Develop into osteoblasts	Endosteum, cellular layer of the periosteum
Osteoblasts	Bone formation	Endosteum, cellular layer of the periosteum, growing portions of bone
Osteocytes	Maintain mineral concentration of matrix	Entrapped in matrix
Osteoclasts	Bone resorption	Endosteum, cellular layer of the periosteum, at sites of old, injured, or unneeded bone

Structure of Cartilage:

Cartilage is a flexible connective tissue that differs from bone in several ways. For one, the primary cell types are chondrocytes as opposed to osteocytes. Chondrocytes are first chondroblast cells that produce the collagen extracellular matrix (ECM) and then get caught in the matrix. They lie in spaces called lacunae with up to eight chondrocytes located in each.

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Chondrocytes rely on diffusion to obtain nutrients as, unlike bone, cartilage is avascular, meaning there are no vessels to carry blood to cartilage tissue. This lack of blood supply causes cartilage to heal very slowly compared with bone.

The base substance of cartilage is chondroitin sulfate, and the microarchitecture is substantially less organized than in bone. The cartilage fibrous sheath is called the perichondrium. The division of cells within cartilage occurs very slowly, and thus growth in cartilage is usually not based on an increase in size or mass of the cartilage itself.

Articular cartilage function is dependent on the molecular composition of its ECM, which consists mainly of proteoglycans and collagens. The remodeling of cartilage is predominantly affected by changes and rearrangements of the collagen matrix, which responds to tensile and compressive forces experienced by the cartilage.

Types of Cartilage:

There are three major types of cartilage: hyaline cartilage, fibrocartilage, and elastic cartilage.

Hyaline Cartilage:

Hyaline cartilage is the most widespread cartilage type and, in adults, it forms the articular surfaces of long bones, the rib tips, the rings of the trachea, and parts of the skull. (This type of cartilage is predominately collagen (yet with few collagen fibers), and its name refers to its glassy appearance.)

In the embryo, bones form first as hyaline cartilage before ossifying as development progresses.

Hyaline cartilage is covered externally by a fibrous membrane, called the perichondrium, except at the articular ends of bones; it also occurs under the skin (for instance, ears and nose).

2 (Hyaline cartilage is found on many joint surfaces. It contains no nerves or blood vessels, and its structure is relatively simple.) 2

(If a thin slice of cartilage is examined under the microscope, it will be found to consist of cells of a rounded or bluntly angular form, lying in groups of two or more in a granular or almost homogeneous matrix.) These cells have generally straight outlines where they are in contact with each other, with the rest of their circumference rounded.

They consist of translucent protoplasm in which fine interlacing filaments and minute granules are sometimes present. Embedded in this are one or two round nuclei with the usual intranuclear network.

Fibrocartilage:

1 (Fibrous cartilage has lots of collagen fibers (Type I and Type II), and it tends to grade into dense tendon and ligament tissue.) White fibrocartilage consists of a mixture of white fibrous tissue and cartilaginous tissue in various proportions.)

It owes its flexibility and toughness to the fibrous tissue, and its elasticity to the cartilaginous tissue. It is the only type of cartilage that contains type I collagen in addition to the normal type II.

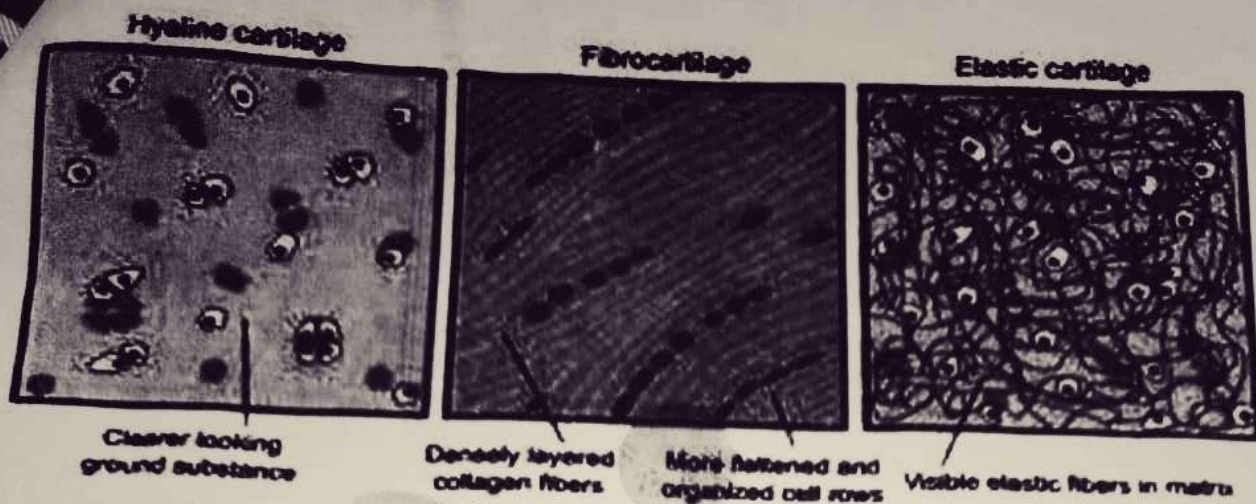
2 (Fibrocartilage is found in the pubic symphysis, the annulus fibrosus of intervertebral discs, menisci, and the temporal mandibular joint.) 2

Elastic Cartilage:

1 (Elastic or yellow cartilage contains elastic fiber networks and collagen fibers. The principal protein is elastin.)

Elastic cartilage is histologically similar to hyaline cartilage but contains many yellow elastic fibers lying in a solid matrix. These fibers form bundles that appear dark under a microscope. They give elastic cartilage great flexibility so it can withstand repeated bending.

Chondrocytes lie between the fibers. (Elastic cartilage is found in the epiglottis (part of the larynx) and the pinnae (the external ear flaps of many mammals).) 2



DIVISION OF HUMAN SKELETON

There are two types of skeleton in man.

- Axial skeleton
- Appendicular skeleton

1. AXIAL SKELETON:

The axial skeleton is **more** of protective nature it consists of following bones.

i. SKULL:

- Skull contains cranium which protects the brain, it is made up of the **8 bones**, these are the occipital bone, two temporal bones, two parietal bones, the sphenoid, ethmoid and frontal bones.
- The second part consists of facial bones which contain **14 bones**, These are the vomer, two inferior nasal conchae, two nasal bones, two maxilla, the mandible, two palatine bones, two zygomatic bones, and two lacrimal bones.
- The **6 auditory ossicles** (little bones) are the malleus, incus and stapes in each ear. The function of the auditory ossicles is to transfer and amplify air vibrations into the inner ear to be processed as sound.
- A throat bone known as **Hyoid bone** is U shaped and provides attachment for tongue and muscles of oral cavity.

ii. VERTEBRAL COLUMN:

- The spinal cord is protected in a hollow spine (back bone) called vertebral column.
- It consists of **33 bones (after fusion 26)** called vertebrae.
- Divided into five regions:
 - 7 cervical, in the neck,
 - 12 thoracic in the chest, which articulates with the ribs,
 - 5 lumbar, in the lower back, more robust than the other vertebrae
 - 5 sacral, often fused to form a sacrum, which articulates with the pelvic girdle

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➤ 4 coccygeal, in the tail.

- The atlas and axis vertebrae, the top two cervicals, form a freely movable joint with the skull.
- The vertebrae are supported by cartilaginous disc, which protect them from grinding.

iii. RIBCAGE:

- The rib contain 12 pairs of bone.
- The ribcage encloses the chest cavity and protects the heart and lungs.
- Ribs are semicircular bones attached on their dorsal side with vertebrae and on their ventral side with the sternum.
- The upper seven pairs of ribs are known as true ribs because their costal cartilages articulate anteriorly directly with the sternum. The lower five pairs of ribs (ribs eight to twelve) are known as false ribs because their costal cartilages do not articulate directly with the sternum.
- The lower two pairs of ribs are also called floating ribs and are not connected to sternum.

2. APPENDICULAR SKELETON:

It consists of 126 bones comprising of arms, legs and pectoral and pelvic girdles.

PECTORAL GIRDLE:

It is found at the shoulder region and consists of two important bones called as scapula and clavicle on both sides

UPPER LIMB:

In the fore arm humerus is attached to the scapula by ball and socket joints. It contains 30 bones (60 in both upper limbs), the bones below humerus are radius and ulna below them are 8 carpals, 5 meta-carpals and 14 small bones called phalanges arranged in 5 rows.

PELVIC GIRDLE:

The hip bone or the pelvic girdle comprise three bones – ilium, ischium and the pubic bone. Pelvic girdle is made up of a pair of single bones called coxal bone (hip bone), which are strongly joined to the sacrum of the vertebral column.

Ilium- Ilium is the largest part of the coxal bone. It is the upper part and has a fan-like structure. It is attached to the sacrum of the vertebral column strongly by the sacroiliac joint.

Ischium- It makes the posterior part of the pelvic girdle below the ilium. It provides support while sitting.

Pubis- It makes the lower anterior part of the pelvic girdle.

Pubic symphysis- It is a joint between the pubis part of the two hip bones in the middle portion. It contains fibrous cartilage.

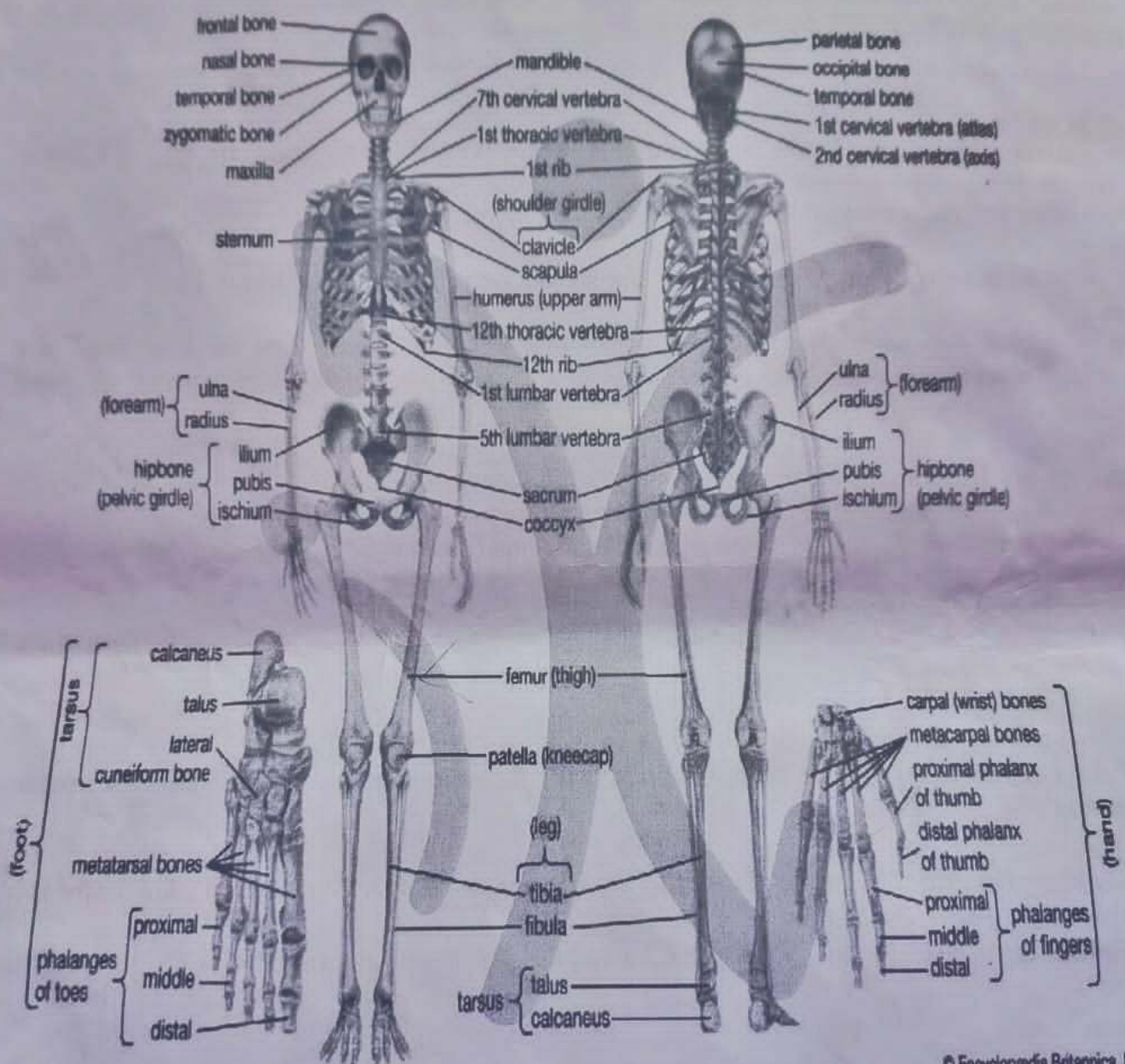
The main function of the hip joint is to support the weight of the body dynamically as it promotes load and force transmission to the lower parts from the axial skeleton

the pelvis by Ball and socket joint. Below it there is fibula and femur, patella (kneecap) 7 tibia

LOWER LIMB:

Lower limb consists of **30 bones (60 in both lower limbs)**

Femur the largest bone of the leg is attached to pelvis. Below it there are **tibia** and **fibula**, 7 tarsals, 5 meta-tarsals and 14 phalanges.



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Figure 3 – Human Skeleton

JOINTS

A joint is defined as a **connection** between two bones in the skeletal system. Joints can be **classified** by the type of the tissue present (fibrous, cartilaginous or synovial), or by the degree of movement permitted (synarthrosis, amphiarthrosis or diarthrosis).

Classification by types of tissues	Classification by degree of tissues ^{movement}
Fibrous – bones connected by fibrous tissue. Cartilaginous – bones connected by cartilage. Synovial – articulating surfaces enclosed within fluid-filled joint capsule.	Synarthrosis – immovable. Amphiarthrosis – slightly moveable. Diarthrosis – freely moveable.

Fibrous Joints

A fibrous joint is where the bones are bound by a tough, fibrous tissue. These are typically joints that require strength and stability over range of movement.

Fibrous joints can be further sub-classified into **sutures**, gomphoses and syndesmoses.

Sutures

Sutures are immovable joints (**synarthrosis**), and are only found **between the** flat, plate-like bones of the skull.

There is limited movement until about 20 years of age, **after** which they become fixed and immobile. They are most important in **birth**, as at that **stage** the joints are not fused, allowing deformation of the skull as it passes through the birth canal.

Gomphoses

Gomphoses are also immovable joints. They are found where the teeth articulate with their sockets in the maxilla (upper teeth) or the mandible (lower teeth).

The tooth is bound into its socket by the strong periodontal ligament.

Syndesmoses

Syndesmoses are slightly movable joints (amphiarthroses).

They are comprised of bones held together by an interosseous membrane. The middle radioulnar joint and middle tibiofibular joint are examples of a syndesmosis joint.

Cartilaginous

In a cartilaginous joint, the bones are united by fibrocartilage or hyaline cartilage.

There are two main types: synchondroses (primary cartilaginous) and symphyses (secondary cartilaginous).

Synchondroses

In a synchondrosis, the bones are connected by hyaline cartilage. These joints are immovable (synarthrosis).

An example of a synchondrosis is the joint between the diaphysis and epiphysis of a growing long bone.

Symphyses

Symphysial joints are where the bones are united by a layer of fibrocartilage. They are slightly movable (amphiarthrosis).

Examples include the pubic symphysis, and the joints between vertebral bodies.

Synovial

A synovial joint is defined by the presence of a fluid-filled joint cavity contained within a fibrous capsule.

They are freely movable (diarthrosis) and are the most common type of joint found in the body.

Synovial joints can be sub-classified into several different types, depending on the shape of their articular surfaces and the movements permitted:

Slide one over another
Sabular

Hinge – permits movement in one plane – usually flexion and extension.
E.g. elbow joint, ankle joint, knee joint.

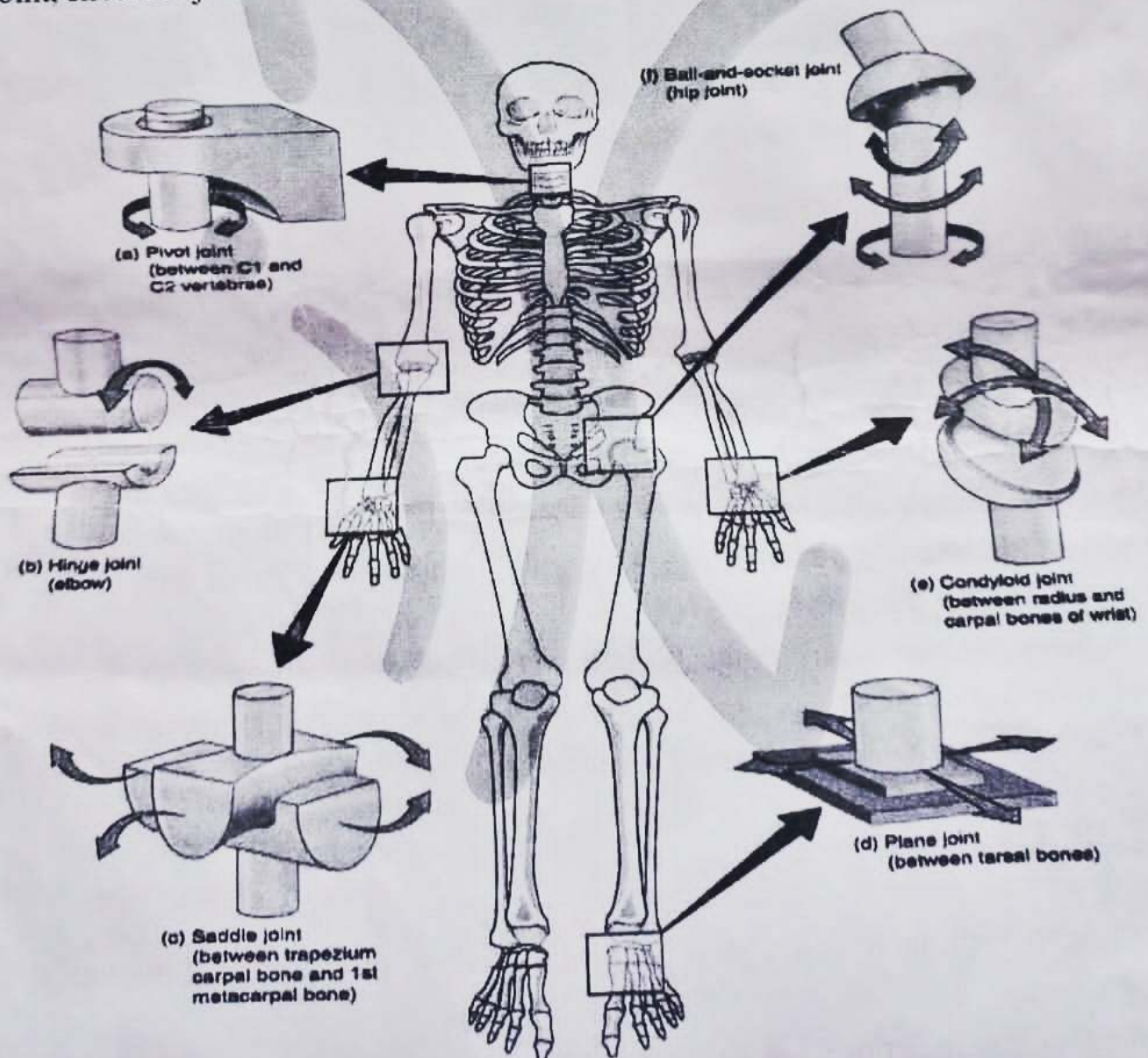
Saddle – named due to its resemblance to a saddle on a horse's back. It is characterised by opposing articular surfaces with a reciprocal concave-convex shape.
E.g. carpometacarpal joints.

Plane – the articular surfaces are relatively flat, allowing the bones to glide over one another.
E.g. acromioclavicular joint, subtalar joint.

Pivot – allows for rotation only. It is formed by a central bony pivot, which is surrounded by a bony-ligamentous ring
E.g. proximal and distal radioulnar joints, atlantoaxial joint.

Condylloid – contains a convex surface which articulates with a concave elliptical cavity. They are also known as ellipsoid joints.
E.g. wrist joint, metacarpophalangeal joint, metatarsophalangeal joint.

Ball and Socket – where the ball-shaped surface of one rounded bone fits into the cup-like depression of another bone. It permits free movement in numerous axes.
E.g. hip joint, shoulder joint.



COMMON SKELETAL RELATED DISORDERS

1. DISC SLIP:

Reason:

- The back bone of our body consists of many vertebrae.
- These are provided with intervertebral discs which act as **shock absorber**.
- They provide mobility to vertebrae and act as safety devices to avoid grinding hence wearing of successive vertebrae.
- Inner cushion like portion of each disc present between adjacent vertebrae is soft and jelly like whereas outer ring is strong and **cartilaginous** and holds the successive vertebrae together.
- If unfortunately due to physical trauma the cartilaginous ring of a disc ruptures and displaces it is called as disc slip.

Clinical effects:

The protrusion presses upon a spinal nerve, emerging from the spinal cord in between the vertebrae causing severe pain and inability to move.

Precaution and treatments:

- As a result of disc slip and the person should use hard bed and should take rest for a long time.
- Pain killer medicines help repair the damage.

2. SPONDYLOSIS:

Reason:

- This is the disorder between two vertebrae in which the space between them become narrow.
- It mostly takes place in the vertebrae of neck (cervical vertebrae).

Effects:

Due to Spondylitis, the nerves emerging from the spinal cord are pressed; it results in the severe pain in the neck, shoulder and upper parts of arm.

Precaution and Treatment:

- In this condition a hard collar is used around the neck which keeps the affected vertebrae, a bit apart and hence the pressure on the nerves is released.
- Pain killer medicines are used to control the disorder.

3. ARTHRITIS:

Disorder:

It is the condition in which a joint becomes swollen, painful and immovable.

Types

The most common types of arthritis include:

- **Osteoarthritis**, or "wear and tear" arthritis, which develops when joint cartilage breaks down from repeated stress. It's the most common form of arthritis.
- **Ankylosing spondylitis**, or arthritis of the spine (usually your lower back).

- **Juvenile arthritis (JA)**, a disorder where the immune system attacks the tissue around joints. JA typically affects children 16 or younger.
- **Gout**, a disease that causes hard crystals of uric acid to form in your joints.
- **Psoriatic arthritis**, joint inflammation that develops in people with **psoriasis** (autoimmune disorder that causes skin irritation).
- **Rheumatoid arthritis**, a disease that causes the immune system to attack synovial membranes in your joints.

Clinical Effects:

- In arthritis, Calcium may be deposited between two bones, due to which defect causes in the soft and elastic cartilage.
- It becomes hard, joint becomes stiff, joints cannot move in a proper way and person suffer badly.

Treatment:

- Knee joint and stiff hip joint can be replaced by artificial rubber or plastic joint.
- Sometimes it is treated by medicines and physiotherapy.

4. SCIATICA:

Reason:

It is the severe pain of hind limb, which occurs when a nerve of **sciatic plexus** is being pressed. may be due to a disc slip in the lower abdomen, a trauma or due to a damage to a branch of **sciatic nerve** by the needle of a syringe while an injection is administer in the iliac vein.

Effects:

- It makes the leg highly painful and virtually immovable.

Treatment:

- The treatment of sciatica is very slow and prolong. There is no permanent treatment of this disorder.

BONE FRACTURE :

- A fracture is a break or a crack in a bone.
- A fracture occurs when force exerted against a bone is stronger than the bone can structurally withstand.
- The most common sites for bone fractures are the wrist, ankle and hip.
- Treatment includes immobilising the bone with a plaster cast, or surgically inserting metal rods or plates to hold the bone pieces together.

Different types of fracture include:

1. **Closed (simple) fracture** – the broken bone has not pierced the skin.
2. **Open (compound) fracture** – the broken bone juts out through the skin, or a wound leads to the fracture site. Infection and external bleeding are more likely.

3. **Complicated fracture** – structures surrounding the fracture are injured. There may be damage to the veins, arteries or nerves, and there may also be injury to the lining of the bone (the periosteum).

HEALING OF BONE FRACTURE :

A fractured or broken bone undergoes repair through four stages:

1 - Hematoma formation:

Blood vessels in the broken bone tear and hemorrhage, resulting in the formation of clotted blood, or a hematoma, at the site of the break. The severed blood vessels at the broken ends of the bone are sealed by the clotting process. Bone cells deprived of nutrients begin to die.

2 - Bone generation: Within days of the fracture, capillaries grow into the hematoma, while phagocytic cells begin to clear away the dead cells. Though fragments of the blood clot may remain, fibroblasts and osteoblasts enter the area and begin to reform bone. Fibroblasts produce collagen fibers that connect the broken bone ends, while osteoblasts start to form spongy bone. The repair tissue between the broken bone ends, the fibrocartilaginous callus, is composed of both hyaline and fibrocartilage. Some bone spicules may also appear at this point.

3 - Bony callous formation: The fibrocartilaginous callus is converted into a bony callus of spongy bone. It takes about two months for the broken bone ends to be firmly joined together after the fracture. This is similar to the endochondral formation of bone when cartilage becomes ossified; osteoblasts, osteoclasts, and bone matrix are present.

4 - Bone remodeling: The bony callus is then remodelled by osteoclasts and osteoblasts, with excess material on the exterior of the bone and within the medullary cavity being removed. Compact bone is added to create bone tissue that is similar to the original, unbroken bone. This remodeling can take many months; the bone may remain uneven for years.

JOINT INJURIES

SPRAIN

A sprain is an injury of a particular joint, such as a knee sprain or an elbow sprain. These sprains occur when a ligament (the tissues that connect different bones within the body) is overstretched. Sprains can range between being very mild to severe. Mild sprains require little more than adequate rest and routine icing, but more serious sprains can leave the joint unstable and prone to an even worse injury. In extreme cases, the joint's ligaments may tear completely away from the bone.

All sprain cases will result in similar symptoms, including:

- Swelling
- Pain near the joint
- Bruising

DISLOCATION

- As the name suggests, a dislocation describes an event in which a joint has become displaced from its normal position. This type of injury tends to be more painful than sprains or strains because they inflict pain when the joint becomes dislocated and is put back into place.
- Individuals should never attempt to relocate their dislocated joint and are highly encouraged to seek medical attention immediately after sustaining a dislocation. This kind of injury will not heal on its own. A dislocation will require repositioning before the recovery process can begin.
- The most frequent joints that suffer from dislocations are:
 - Shoulders
 - Ankles
 - Knees
 - Elbows
 - Fingers
 - Hips

MUSCULAR SYSTEM

Introduction:

- In the body of vertebrates all muscles forms the muscular system.
- Muscles are made up of muscular tissue.
- A muscular tissue is a group of muscle cells.
- The muscles cells are specialized to perform one unique function to generate a pulling force that is they shorten or contract.
- This contraction force or pulling force helps to perform different functions of the body.
- The muscular system works with the skeleton and taken part in the movement of body organs.
- There are more than 600 muscles in a human body and almost half of a body weight is due to muscles.
- A man and woman both have the same no. of bones and muscles.

Functions of Muscular System:

- Muscles helps in the movement of eyelids, tongue food in the alimentary canal.
- They are involved in the working of a heart and circulation of internal fluid.
- They take part in the excretion of harmful substances.
- They help in the secretion of important substances from the glands.
- They take part in the exchange of gases.

Types of Muscles:

There are three types of muscles

- Smooth muscles
- Cardiac Muscles
- Skeletal muscles

1. SMOOTH MUSCLES:

Structure:

- Smooth muscles are structurally the simplest of all muscles types.
- They consists of long spindle shaped i.e. broad in the middle and taper towards the end.
- They are uninucleated.
- Each muscle has cytoplasm called sarcoplasm.
- Around the nucleus the cytoplasm is granular while its remaining part contains elongated fiber, known as myofibrils.

Location:

- These muscles are found in the blood vessels, digestive tract and many other organs.

Control System:

- Smooth muscles are involuntary and have the ability of contraction.
- Their function is controlled by autonomic nervous system.
- They contract more slowly than skeletal muscles, but it can be prolonged for long period of time.

Functions:

- Smooth muscles push the food downward in the alimentary canal.
- They help in the excretion of urine from the urinary bladder.
- These muscles control the size of blood vessels.
- They also control the size of pupil of eye for the entrance of light in sufficient amount.
- These muscles also control the state of erection of hairs on the skin.

2. SKELETAL MUSCLES:

Association:

- The muscles are attached to the bones of the skeleton and enables the skeleton to move. Hence they are called as skeletal muscles.

Appearance:

- Under the microscope skeleton muscles look striped they are called as striated muscles.

Composition:

- These muscles are made up of protein filaments.

Nature:

- They have light and dark bands and can be controlled by one's will, so they are also known as striated voluntary muscles.

Power of Action:

- These muscles show contraction when they receive message from central nervous system. After contraction they rapidly come back to their normal position.

Working Manner:

- Some skeletal muscles are directly attached to the outer covering of the bones whereas a few terminate into tough non elastic tissue called tendon which connects muscles to the bones.
- The bones are attached to each other by a special tissue called ligaments.
- When muscles contracts tendon are stretched and bone moves in this way contraction of muscles help in the movement of bones.
- Muscles must attach at two joints for movement to occur.
- The attachment to the stationary (non-moveable) part called origin and moveable part is called insertion.

3. CARDIAC MUSCLES:**Introduction:**

- Cardiac muscles are so called because they only form the muscular walls of heart of man and most of the vertebrates.

Structure:

- They are uni-nucleated or bi-nucleated and branched to create a meshwork of contractile tissue connected by Intercalated Discs hence their fibers cannot be separated.

Power of Action:

- Heart walls can withstand high pressure without any danger of being fatigued or damaged. As they form a meshwork of the cardiac muscles fibers when contract the entire chamber of heart squeezes to maintain the flow of blood.

Composition:

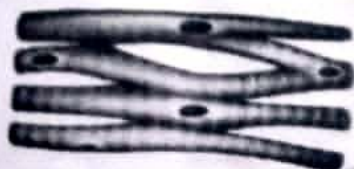
- They contain myosin and actin proteins. Cardiac muscles have more mitochondria to maintain a continuous supply of energy to the tissue of heart.

Control System:

- They contract and relax in a continuous rhythmic pattern called heartbeat.
- It is regulated by Sino-atrial node (SAN) it is termed as pace maker.
- The wave of impulse initiated by pace maker travels the entire heart without involving nerves.

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Types of Muscle



Cardiac muscle



Skeletal muscle



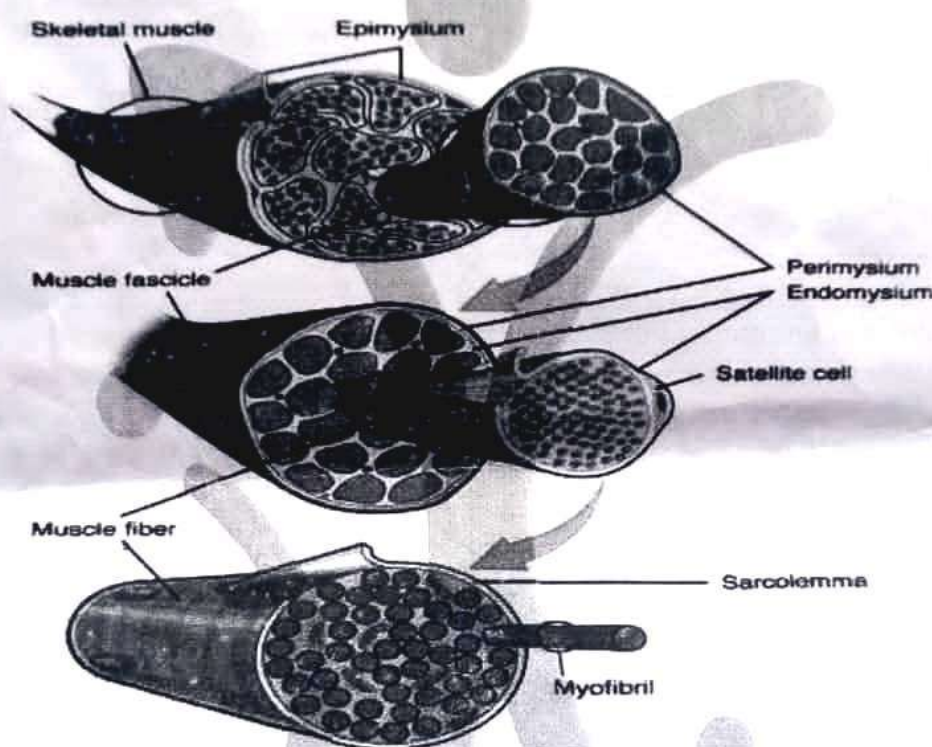
Smooth muscle

Smooth muscles	Cardiac muscles	Skeletal muscles
They do not show striations	They show striations	They show striations
These muscles are spindle shaped	These muscles are cylindrically branched	These are cylindrical and unbranched
Present in hollow, visceral organs	Present in heart	Attached with skeleton
They are involuntary in action	They are involuntary in action	They are voluntary in action
They do not make cross bridges during movement	Cross bridges are formed	Cross bridges are formed
Instead of troponin, they contain calmodulin in actin filaments	They have troponin in actin filament	They have troponin in actin filament

STRUCTURE OF SKELETAL MUSCLES

- Each skeletal muscle has three layers of connective tissue (called "mysia") that enclose it and provide structure to the muscle as a whole, and also compartmentalize the muscle fibers within the muscle. Each muscle is wrapped in a sheath of dense, irregular connective tissue called the epimysium, which allows a muscle to contract and move powerfully while maintaining its structural integrity. The epimysium also separates muscle from other tissues and organs in the area, allowing the muscle to move independently.

- Inside each skeletal muscle, muscle fibers are organized into bundles, called a fascicle, by a middle layer of dense irregular connective tissue called the perimysium. This fascicular organization is common in muscles of the limbs; it allows the nervous system to trigger a specific movement of a muscle by activating a subset of muscle fibers within a bundle, or fascicle, of the muscle.
- Inside each fascicle, each muscle fiber is encased in a thin areolar connective tissue layer of collagen and reticular fibers called the endomysium. The endomysium contains the extracellular fluid and nutrients to support the muscle fiber. These nutrients are supplied via blood capillaries located in the endomysium.



ULTRA STRUCTURE OF SKELETAL MUSCLE FIBRES

- Each skeletal muscle consists of numerous **elongated** and **parallel** fibre like structure called muscle fiber.
- Each muscle fiber is a complete cell, it contains cytoplasm called **sarcoplasm** which is surrounded by an outer membrane called as **sarcolemma**.
- In sarcoplasm many nuclei are present.
- The specialized site of the skeletal muscle where a motor neuron's axon terminal meets the muscle fiber is called the neuromuscular junction (NMJ). This is where the muscle fiber first responds to signaling by the motor neuron. The sarcolemma (plasma membrane) in the region of a neuromuscular junction, known as the motor end-plate, contains special transmembrane proteins that receive a signal from the motor neuron and begin the electrical signal in the muscle fiber

Composition of Muscles Fibers:

- Each of muscle fiber is itself composed of many individual ultra-microscopic contractile **myofibrils**.
- The fibrils are made up of a linear series of **sarcomeres**, which are the actual unit of muscle tissues that contract.
- Surrounding each myofibril is the sarcoplasmic reticulum (SR), the specialized smooth endoplasmic reticulum of muscle fibers, which stores, releases, and retrieves calcium ions (Ca^{++})
- The signal from a motor neuron actually stimulates the release of calcium ions (Ca^{++}) from its storage in the cell's **SR**. For the action potential to reach the membrane of the SR from the motor end plate, there are periodic invaginations in the sarcolemma, called T-tubules ("T" stands for "transverse")

Separation of Sarcomeres:

- Adjacent sarcomeres are separated from one another by **dark Z line**. (Zwischen)
- Between the Z lines are **dark and light zones** which make up a sarcomere.
- Due to these zones the skeletal muscles are called striated muscles.

> MICROFILAMENTS AND ITS TYPES:

- Myofibrils is composed of smallest parts the microfilaments more simply called filaments. These filaments are of two types thick and thin.

i. Thick Filament:

They are made up of protein **myosin**. Myosin myofilaments have two specific regions, the tail and the heads. Connecting these two regions is a flexible hinge region. The tails function to cluster many myosin myofilaments together, while the heads take an active role in the muscle contraction process. The heads contain an actin-binding site, as well as an ATP binding site.

ii. Thin Filament:

Six thin filaments surrounds the myosin filament. Thin filaments are made up of **actin, tropomyosin and troponin**. The troponin protein complex consists of three polypeptides. Troponin I (TnI) binds to actin, troponin T (TnT) binds to tropomyosin, and troponin C (TnC) binds to calcium ions. Troponin and tropomyosin run along the actin filaments and control when the actin binding sites will be exposed for binding to myosin.

> OVERLAPPING PATTERN OF THESE THIN AND THICK FILAMENTS:

The dark and light bands of the fibers results from a regular overlapping pattern of these thin and thick filaments.

I-bands: (Isotropic)

- Each sarcomere has a pair of lightly staining I-bands located at the outer edges of sarcomere.
- It contains **only the actin** filaments.

A-bands: (Anisotropic)

- The more darkly staining A-bands located between the outer sides of I-bands are called A-bands.
- The part A-band on either side of H-Zone represents region of overlapping and contain both thin and thick filaments hence look darker.

H-bands: (Helle)

- A lightly staining H-zone is located in the center of A-band. It has only thick filaments.
- The thick filaments have cross bridges which are attached to the actin filaments during muscle contraction.

MECHANISM OF CONTRACTION OF SKELETAL MUSCLES**Introduction:**

- Whenever a muscle works it contracts and due to contraction it becomes shorter and thicker.
- This contraction of muscles has been explained by Huxley.
- According to him when a muscles contracts the sarcomeres shorten and the width of dark and light bands changes. He proposed sliding filament theory of muscles contraction.

Explanation:

- According to sliding theory of filament the thin and thick filaments of a muscle fibers move together by sliding over each other.
- The sliding over the filaments is the reason that the muscles get shorter and thicker.

Cross Bridge Cycle

- When a nerve signal reaches the muscle cell, calcium is released from the sarcoplasmic reticulum surrounding the myofibrils. Calcium causes a conformational change in the tropomyosin molecule which shift in position to expose the binding sites of the actin proteins.
- The myosin heads bind to the binding sites of the actin proteins, to form a cross-bridge as the inorganic phosphate is released.
- ADP is released which causes initiation of the power stroke, where the thin filament gets pulled closer toward the midline of a sarcomere.
- A new ATP molecule binds to the myosin head causing the separation of the actin-myosin cross-bridge. The ATP is subsequently hydrolyzed to ADP and inorganic Phosphate (step 1) and the energy transferred from the ATP to the myosin head causes it to "cock" back like the trigger of a gun.
- This contraction cycle will continue until the nerve signal stops, and calcium is reabsorbed back into the sarcoplasmic reticulum, which causes the tropomyosin molecules to cover the actin binding sites, stopping the myosin to form new cross-bridges.

ARRANGEMENT OF SKELETAL MUSCLES AT KNEE JOINT

ANTAGONISTIC MUSCLES:

Introduction:

The muscles work in pair against each other in opposite way for the movement of bone, they are called antagonistic muscles.

Arrangement:

The skeletal muscles must work in opposing pairs because one set of muscle pulls on a tendon to bend a joint, another set of muscles must pull on a different tendon to straighten the same joint. Combination of various muscles permits the controlled movement of a joint in several directions.

AT KNEE JOINT

- The quads and hamstrings control the extension and contraction of the knees, which are an integral part of some of the most basic things we can do, such as walking or sitting down.
- When our legs are relaxed (such as when we're in a more natural standing position), the quads function as the agonist muscle, in that they contract and tense, while the hamstring is the antagonist muscle, meaning it's relaxed.
- However, when the leg is bent (when you're crouched or squatting, for instance), these roles are switched - the hamstring is now the agonist muscle, whereas the quads are antagonist muscles in this scenario.

MUSCLE FATIGUE

Definition:

When the muscle become functionless and so tired due to accumulation of **lactic acid** it is called muscle fatigue.

Reason:

- During a heavy or prolong exercise when enough energy is not available muscle cells receive supply of ATP through fermentation also called **anaerobic respiration**.
- During fermentation glucose is converted in lactic acid instead of CO_2 and H_2O as the case is with normal aerobic respiration.
- This accumulation of lactic acid in muscles produces a tiring condition of the muscles called fatigue.

Recovery:

- When this heavy exercise stops heavy breathing continues to supply the excess oxygen to the fatigued tissues which now break lactic acid into water and CO_2 or convert it into glycogen.
- By this process the muscles again become normal and perform their function in regular manner.

Oxygen Debt:

- The amount of oxygen which is used to remove lactic acid from fatigued muscles is called oxygen debt, i.e. loss of oxygen.
- The amount of lactic acid in the muscles indicates the amount of oxygen debt.
- Active and constant use of muscles makes them stronger whereas complete immobilization makes them weaker and subject to fatigue soon.

CRAMPS**Introduction:**

- A cramp is a sudden, and involuntary muscle contraction or over-shortening.
- It is generally temporary and non-damaging, they can cause pain, and a paralysis-like immobility of the affected muscle(s).

Time Duration:

- Onset is usually sudden, and it resolves on its own over a period of several seconds, minutes, or hours.

Cause:

- Cramps may occur in a skeletal muscle or smooth muscle. Skeletal muscle cramps may be caused by any combination of muscle fatigue, a lack of electrolytes (e.g., low sodium, low potassium, or low magnesium).
- Cramps of smooth muscle may be due to menstruation or gastroenteritis.

Treatment:

- Stretching, massage and drinking plenty of fluid, such as water, may be helpful in treating simple muscle cramps.

Medication:

- Vitamin B complex and calcium channel blockers may be effective for muscle cramps.

TETANY**Definition:**

When skeletal muscles start sudden contraction involuntarily without any control it is called as tetany.

Causes:

Tetany is a disease caused by low calcium in the blood.

Effect:

- It causes stimulation in neuron and by their influence the muscles contract before gaining the normal position of actin and myosin filaments therefore it is called as abnormal function.
- In tetany there is a continuous contraction of muscle fibers.
- Due to this continuous contraction the Calcium ions cannot be separated from the sarcoplasm of muscles and continue contraction becomes very rapid.

Symptoms:

- It increases the excitability of neuron and results in the loss of sensation.
- Muscle twitches and convulsion occurs.
- If untreated the symptoms progresses.
- E.g. spasm of larynx respiratory paralysis and ultimately death occur.

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